

LOVIISA ORRENGRUND, Finland

WMO: 029920

Lat: 60.275N

Lon: 26.444E

Elev: 4

StdP: 101.28

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.2	-15.4	-20.5	0.6	-18.0	-17.5	0.8	-15.1	16.0	-2.2	14.9	-1.9	5.8	10	0.784

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.2	24.2	20.6	22.8	19.5	21.3	18.2	21.1	23.5	20.0	22.6	18.6	20.8	4.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.1	14.8	23.0	18.9	13.7	22.0	17.5	12.5	20.3	61.1	23.7	57.5	22.6	52.7	21.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.3	13.8	12.4	DB	-18.5	25.0	5.0	2.4	-22.1	26.7	-25.0	28.2	-27.9	29.5	-31.5	31.3	(4)
(5)			WB	-18.7	21.0	5.0	1.8	-22.3	22.3	-25.3	23.3	-28.1	24.3	-31.7	25.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-3.6	-4.6	-2.0	3.2	9.2	13.6	17.8	17.1	13.1	7.6	3.4	0.4	(6)	
	DBStd	8.43	5.16	5.57	4.09	2.48	3.37	2.57	2.93	2.45	2.56	3.05	3.42	3.96	(7)	
	HDD10.0	2054	421	410	372	206	56	3	0	0	6	86	199	298	(8)	
	HDD18.3	4428	679	643	630	455	282	143	46	53	157	334	449	557	(9)	
	CDD10.0	715	0	0	0	0	32	111	242	222	99	10	0	0	(10)	
	CDD18.3	47	0	0	0	0	0	1	30	16	0	0	0	0	(11)	
	CDH23.3	79	0	0	0	0	0	1	64	14	0	0	0	0	(12)	
	CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0	(13)	
(14) Wind	WSAvg	6.5	7.4	6.4	6.4	5.6	5.4	5.5	5.1	5.6	6.6	7.4	8.2	8.5	(14)	
(15) Precipitation	PrecAvg	616	52	43	36	33	37	60	60	62	55	66	62	50	(15)	
	PrecMax	757	100	108	70	73	80	128	132	130	108	189	151	111	(16)	
	PrecMin	458	3	0	5	10	5	6	7	0	7	25	0	6	(17)	
	PrecStd	77	25	28	16	16	24	32	37	34	33	37	33	24	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	3.4	5.5	12.3	19.7	22.2	26.6	25.0	19.0	13.9	9.8	7.6	(19)	
		MCWB	4.0	1.9	3.2	8.2	14.3	17.6	21.0	21.3	16.9	12.6	9.0	6.8	(20)	
	2%	DB	3.9	2.2	3.8	9.6	17.6	20.3	25.0	23.2	18.0	13.1	9.1	6.5	(21)	
		MCWB	3.4	1.5	2.1	6.7	13.2	16.1	20.9	20.0	15.9	12.1	8.4	5.7	(22)	
	5%	DB	3.1	1.7	2.9	7.7	15.9	18.8	23.8	21.9	17.2	12.5	8.3	5.8	(23)	
		MCWB	2.6	1.2	1.7	5.2	12.5	15.2	20.3	18.6	15.2	11.5	7.7	5.1	(24)	
	10%	DB	2.3	1.2	2.2	6.5	14.2	17.4	22.5	20.7	16.5	11.7	7.5	4.9	(25)	
		MCWB	1.8	0.8	1.3	4.4	11.4	14.4	19.5	17.6	14.7	10.7	6.8	4.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.3	2.3	3.7	8.5	15.2	18.8	22.5	21.9	17.5	13.1	9.3	7.1	(27)	
		MCDB	4.6	2.6	4.9	11.5	18.2	21.1	25.0	24.3	18.5	13.5	9.5	7.5	(28)	
	2%	WB	3.6	1.8	2.5	6.9	14.0	16.7	21.5	20.5	16.5	12.4	8.6	6.0	(29)	
		MCDB	3.8	2.1	3.3	9.4	16.7	19.3	24.1	22.8	17.5	13.0	8.9	6.4	(30)	
	5%	WB	2.7	1.3	2.0	5.7	12.8	15.7	20.7	19.2	15.7	11.7	7.8	5.1	(31)	
		MCDB	3.0	1.6	2.6	7.2	15.3	18.2	23.3	21.3	16.8	12.3	8.2	5.6	(32)	
	10%	WB	1.9	0.9	1.4	4.8	11.5	14.7	19.8	18.1	15.1	10.9	6.9	4.3	(33)	
		MCDB	2.3	1.2	2.0	6.0	14.1	17.1	22.5	20.0	16.3	11.6	7.3	4.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.7	3.6	4.1	3.9	4.6	4.5	4.2	3.7	3.1	2.8	2.7	3.1	(35)	
		MCDBR	2.8	2.3	3.6	5.6	6.5	6.0	4.9	4.4	3.5	2.7	2.5	2.9	(36)	
	5% WB	MCWBR	3.0	2.2	2.6	3.3	3.8	3.4	2.9	2.8	2.6	2.7	2.8	3.0	(37)	
		MCWBR	2.7	2.3	3.2	5.0	5.9	5.6	4.6	4.2	3.1	2.7	2.5	2.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.280	0.312	0.334	0.371	0.368	0.365	0.386	0.379	0.353	0.335	0.307	0.385	(40)	
		taud	2.138	2.168	2.212	2.284	2.402	2.460	2.420	2.435	2.491	2.385	2.210	2.601	(41)	
	Edn at Noon		468	665	783	824	858	867	835	811	774	649	431	328	(42)	
		Edh at Noon	43	73	97	109	104	100	102	94	75	60	40	29	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.25	0.85	2.34	3.86	5.30	5.52	5.50	4.20	2.50	1.04	0.28	0.14	(44)	
	RadStd		0.05	0.18	0.30	0.42	0.57	0.38	0.49	0.43	0.30	0.15	0.05	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)		+0.47	N/A	N/A	N/A	N/A	N/A	-155	-165	N/A	N/A

Nomenclature: See separate page